

Online Document for “Symmetry-Aware Learning Method for Warm-Starting Large-Scale Unit Commitment”

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This online document accompanies the main paper and contains the complete fast-ramping formulation and proof, the slow-ramping formulation, and the RSVOR solution derivations cited there. Equation and algorithm numbers continue those used in the main paper. Unless stated otherwise, notation follows the main paper.

APPENDIX A

FAST-RAMPING AGGREGATE FORMULATION AND PROOF

This appendix gives the complete formulation used for the fast-ramping clusters and proves Theorem 1 of the main paper.

A. Aggregate Formulation

For every $A \in \mathcal{A}^F$ and $t \in \mathcal{T}$, the aggregate commitment and minimum up/down time constraints are

$$U_{At} - U_{A,t-1} = V_{At} - W_{At} \quad \forall t, \quad (30)$$

$$\sum_{i=t-TU_A+1}^t V_{Ai} \leq U_{At} \quad \forall t \in \mathcal{T}, \quad (31)$$

$$\sum_{i=t-TD_A+1}^t W_{Ai} \leq K_A - U_{At} \quad \forall t \in \mathcal{T}. \quad (32)$$

For $TU_A \geq 2$, the output is bounded by

$$P_{At} \leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SU_A)V_{At} - (\bar{\Pi}_A - SD_A)W_{A,t+1}. \quad (33)$$

For $TU_A = 1$, this bound is replaced by the pair

$$P_{At} \leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SD_A)W_{A,t+1} - \max\{SD_A - SU_A, 0\}V_{At}, \quad (34)$$

$$P_{At} \leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SU_A)V_{At} - \max\{SU_A - SD_A, 0\}W_{A,t+1}. \quad (35)$$

The aggregate ramping constraints are

$$P_{At} - P_{A,t-1} \leq RU_A U_{A,t-1} + (SU_A - \underline{\Pi}_A)V_{At}, \quad (36)$$

$$P_{A,t-1} - P_{At} \leq RD_A U_{At} + (SD_A - \underline{\Pi}_A)W_{At}. \quad (37)$$

The variable domains are

$$U_{At}, V_{At}, W_{At} \in \{0, \dots, K_A\}, \quad P_{At} \geq 0. \quad (38)$$

The system power balance constraint is:

$$\sum_{A \in \mathcal{G}_A} (\underline{\Pi}_A U_{At} + P_{At}) = \sum_{b \in B} D_{b,t}, \quad \forall t, \quad (39)$$

Define the net injection at bus b as

$$\mathcal{I}_{bt} = \sum_{A \in \mathcal{G}_A} C_{b,A} (\underline{\Pi}_A U_{At} + P_{At}) - D_{b,t}. \quad (40)$$

The transmission constraints are then

$$-F_l^{\max} \leq \sum_{b \in B} H_{l,b}^{\text{PTDF}} \mathcal{I}_{bt} \leq F_l^{\max} \quad \forall l, t. \quad (41)$$

The corresponding objective is

$$\min \sum_{\substack{t \in \mathcal{T} \\ A \in \mathcal{G}_A}} \left[C_A^{NL} U_{At} + C_A^{LV} (\underline{\Pi}_A U_{At} + P_{At}) + C_A^{\text{start}} V_{At} \right]. \quad (42)$$

B. Proof of Theorem 1

Proof: Fix $A \in \mathcal{A}^F$. Its normal ramp limits satisfy $RU_A \geq \bar{\Pi}_A - \underline{\Pi}_A$ and $RD_A \geq \bar{\Pi}_A - \underline{\Pi}_A$.

First, take any feasible unit-level schedule and define the aggregate variables by the aggregation identities in equation (12) of the main paper. The domains in (38) follow immediately from the unit-level domains. Summing the unit-level commitment and minimum-time constraints gives (30)–(32); summing the unit-level ramping constraints gives (36) and (37). When $TU_A \geq 2$, summing the joint startup and shutdown output bound gives (33).

For $TU_A = 1$, the two unit-level output limits in equations (7) and (8) of the main paper imply

$$p_{gt} \leq (\bar{\Pi}_A - \underline{\Pi}_A)u_{gt} - (\bar{\Pi}_A - SU_A)v_{gt} - (\bar{\Pi}_A - SD_A)w_{g,t+1} + (\bar{\Pi}_A - \max\{SU_A, SD_A\})v_{gt}w_{g,t+1}. \quad (43)$$

For binary transition variables, the right-hand side is exactly the minimum of the two individual bounds. Let $m_{At} := \sum_{g \in A} v_{gt}w_{g,t+1}$ denote the number of units that start at t and shut down at $t+1$. Then $m_{At} \leq \min\{V_{At}, W_{A,t+1}\}$, so summation gives

$$P_{At} \leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SU_A)V_{At} - (\bar{\Pi}_A - SD_A)W_{A,t+1} + (\bar{\Pi}_A - \max\{SU_A, SD_A\}) \cdot \min\{V_{At}, W_{A,t+1}\} = (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - \max\{(\bar{\Pi}_A - SD_A)W_{A,t+1} + \max\{SD_A - SU_A, 0\}V_{At}, (\bar{\Pi}_A - SU_A)V_{At} + \max\{SU_A - SD_A, 0\}W_{A,t+1}\}. \quad (44)$$

Thus, (34) and (35) hold. Finally, aggregation preserves total generation:

$$\sum_{g \in A} (\underline{\Pi}_A u_{gt} + p_{gt}) = \underline{\Pi}_A U_{At} + P_{At}. \quad (45)$$

Since units in A affect line flows in the same way, aggregation also preserves line flows. Thus, the power balance and

network constraints remain satisfied, and every feasible unit-level schedule yields a feasible aggregate schedule.

Conversely, consider any aggregate schedule satisfying (30)–(38). The logical and minimum-time constraints give

$$\begin{aligned} V_{At} &\leq K_A - U_{A,t-1} - \sum_{i=t-TD_A+1}^{t-1} W_{Ai}, \\ W_{At} &\leq U_{A,t-1} - \sum_{i=t-TU_A+1}^{t-1} V_{Ai}. \end{aligned} \quad (46)$$

At period t , the first sum counts units that must remain offline to satisfy their minimum down times. The second counts units that must remain online to satisfy their minimum up times. Thus, we can select V_{At} eligible offline units to start up and W_{At} eligible online units to shut down. Repeating this selection period by period and updating each unit's on/off status gives binary variables u_{gt}, v_{gt}, w_{gt} that satisfy the unit-level commitment and minimum-time constraints. Summing these variables over all units in A gives U_{At}, V_{At}, W_{At} , respectively, as required by the first three identities in equation (12) of the main paper.

For $TU_A = 1$, choose units to shut down at $t+1$ from those started at t first, then from the other online units. Every unit started at t can shut down at $t+1$, and $W_{A,t+1} \leq U_{At}$, so this choice gives

$$\sum_{g \in A} v_{gt} w_{g,t+1} = \min\{V_{At}, W_{A,t+1}\}. \quad (47)$$

For this binary schedule, define each unit's output capacity for $TU_A = 1$ by

$$\begin{aligned} \bar{p}_{gt} &= (\bar{\Pi}_A - \underline{\Pi}_A)u_{gt} - (\bar{\Pi}_A - SU_A)v_{gt} - (\bar{\Pi}_A - SD_A)w_{g,t+1} \\ &\quad + (\bar{\Pi}_A - \max\{SU_A, SD_A\})v_{gt}w_{g,t+1}, \end{aligned} \quad (48)$$

and for $TU_A \geq 2$ by

$$\begin{aligned} \bar{p}_{gt} &= (\bar{\Pi}_A - \underline{\Pi}_A)u_{gt} - (\bar{\Pi}_A - SU_A)v_{gt} \\ &\quad - (\bar{\Pi}_A - SD_A)w_{g,t+1}. \end{aligned} \quad (49)$$

For $TU_A \geq 2$, the minimum up time gives $v_{gt}w_{g,t+1} = 0$. In both cases, this capacity equals the unit-level output limit and satisfies $0 \leq \bar{p}_{gt} \leq \bar{\Pi}_A - \underline{\Pi}_A$. By (44), (47), and (33), the total capacity $Q_{At} := \sum_{h \in A} \bar{p}_{ht}$ satisfies $Q_{At} \geq P_{At}$. Distribute the aggregate output in proportion to these capacities:

$$p_{gt} = \begin{cases} \frac{P_{At}\bar{p}_{gt}}{Q_{At}}, & Q_{At} > 0, \\ 0, & Q_{At} = 0. \end{cases} \quad (50)$$

If $Q_{At} = 0$, nonnegativity and $P_{At} \leq Q_{At}$ imply $P_{At} = 0$. Thus, in both cases,

$$0 \leq p_{gt} \leq \bar{p}_{gt}, \quad \sum_{g \in A} p_{gt} = P_{At}. \quad (51)$$

These bounds satisfy equations (7)–(9) of the main paper.

Next, check equations (10) and (11) of the main paper for the four possible status transitions. If a unit stays online, both outputs lie in $[0, \bar{\Pi}_A - \underline{\Pi}_A]$, so

$$\begin{aligned} p_{gt} - p_{g,t-1} &\leq \bar{\Pi}_A - \underline{\Pi}_A \leq RU_A, \\ -p_{gt} + p_{g,t-1} &\leq \bar{\Pi}_A - \underline{\Pi}_A \leq RD_A. \end{aligned} \quad (52)$$

If the unit starts up, $p_{g,t-1} = 0$ and the startup output bound $p_{gt} \leq SU_A - \underline{\Pi}_A$ ensure that the ramp-up constraint holds, while the left-hand side of the ramp-down constraint is nonpositive. If the unit shuts down, $p_{gt} = 0$ and the shutdown output bound $p_{g,t-1} \leq SD_A - \underline{\Pi}_A$ for the previous period ensure that the ramp-down constraint holds, while the left-hand side of the ramp-up constraint is nonpositive. If the unit remains offline, both outputs are zero. Hence, both ramping constraints hold in all four cases. Finally, (45) preserves power balance and line flows. Therefore, every feasible aggregate schedule can be converted into a feasible unit-level schedule.

Since all units in a cluster have the same cost coefficients, their total cost in each period can be expressed using the aggregate variables:

$$\begin{aligned} \sum_{g \in A} \left[C_A^{NL} u_{gt} + C_A^{LV} (\underline{\Pi}_A u_{gt} + p_{gt}) + C_A^{start} v_{gt} \right] \\ = C_A^{NL} U_{At} + C_A^{LV} (\underline{\Pi}_A U_{At} + P_{At}) + C_A^{start} V_{At}. \end{aligned} \quad (53)$$

Summing (53) over periods and clusters shows that the objective value is unchanged. This proves all three claims of Theorem 1 for fast-ramping clusters. ■

APPENDIX B

SLOW-RAMPING CLUSTERED-UC FORMULATION

For $A \in \mathcal{A}^S$, the implementation uses the clustered-UC formulation of [1].

A. Cluster-Level Constraints

For each $A \in \mathcal{A}^S$, the aggregate commitment and minimum up/down time constraints are

$$U_{At} - U_{A,t-1} = V_{At} - W_{At} \quad \forall t, \quad (54)$$

$$\sum_{i=t-TU_A+1}^t V_{Ai} \leq U_{At} \quad \forall t \in [TU_A, T], \quad (55)$$

$$\sum_{i=t-TD_A+1}^t W_{Ai} \leq K_A - U_{At} \quad \forall t \in [TD_A, T]. \quad (56)$$

When $TU_A \geq 2$, the cluster output limit is

$$\begin{aligned} P_{At} &\leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SU_A)V_{At} \\ &\quad - (\bar{\Pi}_A - SD_A)W_{A,t+1}. \end{aligned} \quad (57)$$

When $TU_A = 1$, it is replaced by the pair

$$\begin{aligned} P_{At} &\leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SD_A)W_{A,t+1} \\ &\quad - \max\{SD_A - SU_A, 0\}V_{At}, \end{aligned} \quad (58)$$

$$\begin{aligned} P_{At} &\leq (\bar{\Pi}_A - \underline{\Pi}_A)U_{At} - (\bar{\Pi}_A - SU_A)V_{At} \\ &\quad - \max\{SU_A - SD_A, 0\}W_{A,t+1}. \end{aligned} \quad (59)$$

The variable domains are

$$P_{At} \geq 0, \quad U_{At}, V_{At}, W_{At} \in \{0, \dots, K_A\}. \quad (60)$$

B. Ordered Individual Constraints

The ordered individual commitment variables satisfy:

$$\tilde{u}_{A,k+1,t} \leq \tilde{u}_{Akt} \quad k = 1, \dots, K_A - 1, \quad (61)$$

$$\tilde{u}_{Akt} \in \{0, 1\}, \quad U_{At} = \sum_{k=1}^{K_A} \tilde{u}_{Akt}. \quad (62)$$

Individual and aggregate outputs satisfy

$$0 \leq \tilde{p}_{Akt} \leq (\bar{\Pi}_A - \underline{\Pi}_A) \tilde{u}_{Akt}, \quad (63)$$

$$P_{At} = \sum_{k=1}^{K_A} \tilde{p}_{Akt}.$$

For $TU_A \geq 2$, the individual output limit is given by

$$\tilde{p}_{Akt} \leq (SU_A - \underline{\Pi}_A) \tilde{u}_{Akt} + (\bar{\Pi}_A - SU_A) \tilde{u}_{Ak,t-1}, \quad (64)$$

$$\tilde{p}_{Akt} \leq (SD_A - \underline{\Pi}_A) \tilde{u}_{Akt} + (\bar{\Pi}_A - SD_A) \tilde{u}_{Ak,t+1}. \quad (65)$$

For $TU_A = 1$, (64) and (65) are replaced by

$$\begin{aligned} \tilde{p}_{Akt} &\leq (SU_A - \bar{\Pi}_A + SD_A - \underline{\Pi}_A) \tilde{u}_{Akt} \\ &\quad + (\bar{\Pi}_A - SU_A) \tilde{u}_{Ak,t-1} + (\bar{\Pi}_A - SD_A) \tilde{u}_{Ak,t+1}. \end{aligned} \quad (66)$$

Finally, the individual ramping constraints are

$$\tilde{p}_{Akt} - \tilde{p}_{Ak,t-1} \leq RU_A \tilde{u}_{Akt}, \quad (67)$$

$$-\tilde{p}_{Akt} + \tilde{p}_{Ak,t-1} \leq RD_A \tilde{u}_{Ak,t-1}. \quad (68)$$

APPENDIX C

SOLUTION METHOD FOR RSVOR TRAINING

This appendix gives the solution details omitted from Section III-C1 of the main paper. The iterative procedure follows the difference-of-convex (DC) framework in [2]. Consider the general problem

$$\min_{u \in \Omega} w(u) + \sum_m h_m(g_m(u)), \quad (69)$$

where u is the decision variable, Ω is its feasible set, m indexes the component terms, $w(u)$ and $g_m(u)$ are arbitrary functions, and each DC function can be written as $h_m = h_{m1} - h_{m2}$, with h_{m1} and h_{m2} convex. At iteration t , let q_{mt} be a subgradient of h_{m2} with respect to $g_m(u)$ at $g_m(u_t)$. The concave component is linearized and the next iterate is obtained from

$$\begin{aligned} u_{t+1} = \arg \min_{u \in \Omega} & w(u) + \sum_m h_{m1}(g_m(u)) \\ & - \sum_m q_{mt} g_m(u). \end{aligned} \quad (70)$$

The resulting generic procedure is summarized in Algorithm 4.

Algorithm 4 DC algorithm for problem (69).

Require: $u_1 \in \Omega$ and $t = 1$.

- 1: **while** not converged **do**
 - 2: Compute each subgradient q_{mt} at $g_m(u_t)$.
 - 3: Update u_{t+1} by solving (70).
 - 4: $t \leftarrow t + 1$.
 - 5: **end while**
-

For RSVOR, each GE term defined in equation (17) of the main paper has the DC decomposition

$$\begin{aligned} \min\{\ell(r_{ij}), \ell(s_j)\} &= \ell(r_{ij}) - h_{s_j}(r_{ij}), \\ h_{s_j}(u) &= \max\{s_j - u, 0\}. \end{aligned} \quad (71)$$

Let $\mathcal{I} := \{1, \dots, N\}$ be the training-sample index set. At iteration q , define the indicator matrix $\mathbf{I}^{(q)} = [I_{ij}^{(q)}] \in \{0, 1\}^{N \times K_A}$ by

$$I_{ij}^{(q)} = \mathbb{I}\{r_{ij}^{(q)} < s_j\}, \quad (72)$$

where $\mathbb{I}\{\cdot\}$ is the indicator function. Fix an outer iteration and suppress the superscript q . Linearizing the concave term in (71) and dropping terms that are constant at the current iteration gives the convex subproblem

$$\begin{aligned} \min_{\mathbf{w}, \mathbf{b}} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^N \sum_{j=1}^{K_A} [\ell(r_{ij}) + I_{ij} r_{ij}] \\ \text{s.t.} \quad & b_1 \leq \dots \leq b_{K_A}, \\ & b_{j-1} - b_j \leq s_j, \quad j = 2, \dots, K_A. \end{aligned} \quad (73)$$

where every (i, j) -indexed expression or constraint applies to $i \in \mathcal{I}$ and $j = 1, \dots, K_A$, unless a different range is shown. Before deriving the QP and LP updates, we verify that relaxing the explicit threshold-order constraint does not destroy the ordinal structure of the predictor. The following result establishes this property.

Theorem 2 (Preservation of threshold ordering). Let $(\mathbf{w}^*, \mathbf{b}^*)$ be any optimal solution of the relaxed convex subproblem obtained from (73) by omitting the explicit constraint $b_1 \leq \dots \leq b_{K_A}$ while retaining $s_j \leq 0$ and $b_{j-1} - b_j \leq s_j$ for $j = 2, \dots, K_A$. Then the optimal threshold vector satisfies $b_1^* \leq \dots \leq b_{K_A}^*$.

Proof: For every $j = 2, \dots, K_A$, the retained cutoff-admissibility conditions give

$$b_{j-1}^* - b_j^* \leq s_j \leq 0. \quad (74)$$

Hence, $b_{j-1}^* \leq b_j^*$ for every adjacent pair, which implies $b_1^* \leq \dots \leq b_{K_A}^*$. Thus, removing the explicit ordering constraint neither enlarges the admissible set nor changes the optimal value; no uniqueness assumption on the threshold vector is required. ■

Theorem 2 ensures that the relaxed update cannot produce unordered cutpoints or change the optimal value. Although the explicit ordering inequalities can therefore be omitted, they are retained below to make the ordinal structure transparent. Introducing slack variables $\eta_{ij} \geq 0$ for the hinge terms yields

$$\begin{aligned} \min_{\mathbf{w}, \mathbf{b}, \boldsymbol{\eta}} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^N \sum_{j=1}^{K_A} \eta_{ij} \\ & + C \sum_{i=1}^N \sum_{j=1}^{K_A} I_{ij} y_i^j (\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle - b_j) \\ \text{s.t.} \quad & y_i^j (\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle - b_j) \geq 1 - \eta_{ij}, \\ & \eta_{ij} \geq 0, \quad b_1 \leq \dots \leq b_{K_A}, \\ & b_{j-1} - b_j \leq s_j, \quad j = 2, \dots, K_A. \end{aligned} \quad (75)$$

For fixed $\mathbf{b}$, let α_{ij} and γ_{ij} be the nonnegative multipliers of the hinge and nonnegativity constraints, respectively. The Lagrangian of (75) is

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^N \sum_{j=1}^{K_A} \eta_{ij} \\ & + C \sum_{i=1}^N \sum_{j=1}^{K_A} I_{ij} y_i^j (\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle - b_j) \\ & + \sum_{i=1}^N \sum_{j=1}^{K_A} \alpha_{ij} \left[1 - y_i^j (\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle - b_j) - \eta_{ij} \right] \\ & - \sum_{i=1}^N \sum_{j=1}^{K_A} \gamma_{ij} \eta_{ij}. \end{aligned} \quad (76)$$

The stationarity conditions with respect to $\mathbf{w}$ and η_{ij} are

$$\mathbf{w} - \sum_{i=1}^N \sum_{j=1}^{K_A} \alpha_{ij} y_i^j \phi(\mathbf{x}_i) + C \sum_{i=1}^N \sum_{j=1}^{K_A} I_{ij} y_i^j \phi(\mathbf{x}_i) = 0, \quad (77)$$

$$C - \alpha_{ij} - \gamma_{ij} = 0. \quad (78)$$

Thus, $0 \leq \alpha_{ij} \leq C$. Define $\beta_{ij} = \alpha_{ij} - C I_{ij}$. Eliminating $\mathbf{w}$ and $\boldsymbol{\eta}$ gives the dual QP

$$\begin{aligned} \min_{\boldsymbol{\alpha}} \quad & \frac{1}{2} \sum_{i=1}^N \sum_{i'=1}^N \sum_{j=1}^{K_A} \sum_{j'=1}^{K_A} \beta_{ij} \beta_{i'j'} y_i^j y_{i'}^{j'} K(\mathbf{x}_i, \mathbf{x}_{i'}) \\ & - \sum_{i=1}^N \sum_{j=1}^{K_A} \alpha_{ij} - \sum_{i=1}^N \sum_{j=1}^{K_A} \beta_{ij} y_i^j b_j \\ \text{s.t.} \quad & 0 \leq \alpha_{ij} \leq C, \quad i \in \mathcal{I}, j = 1, \dots, K_A. \end{aligned} \quad (79)$$

After solving (79), the projection vector is recovered from

$$\mathbf{w} = \sum_{i=1}^N \sum_{j=1}^{K_A} \beta_{ij} y_i^j \phi(\mathbf{x}_i). \quad (80)$$

With $\mathbf{w}$ fixed, terms independent of $\mathbf{b}$ can be removed from (75), so the thresholds are updated by the LP

$$\begin{aligned} \min_{\boldsymbol{\eta}, \mathbf{b}} \quad & \sum_{i=1}^N \sum_{j=1}^{K_A} \eta_{ij} - \sum_{i=1}^N \sum_{j=1}^{K_A} I_{ij} y_i^j b_j \\ \text{s.t.} \quad & y_i^j (\langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle - b_j) \geq 1 - \eta_{ij}, \\ & \eta_{ij} \geq 0, \quad b_1 \leq \dots \leq b_{K_A}, \\ & b_{j-1} - b_j \leq s_j, \quad j = 2, \dots, K_A. \end{aligned} \quad (81)$$

The complete RSVOR procedure is given in Algorithm 1 of the main paper. It alternates between the dual QP in (79), the recovery step in (80), the threshold LP in (81), and the indicator update in (72).

REFERENCES

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- [2] H. Xiang, Z. Wu, R. Wang, F. Nie, and X. Li, "Robust support vector ordinal regression," *Information Sciences*, vol. 717, p. 122277, 2025.